CYBERSNOVA
CYBER SECURITY INTERNSHIP

PROJECT 1
NETWORK RECONNAISSANCE & SERVICE ENUMERATION

Reconnaissance Report

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Domain	Cyber Security
Project Level	Beginner
Project	Network Reconnaissance & Service Enumeration
Target	http://testphp.vulnweb.com/
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1. Introduction

Network reconnaissance is an initial phase of a security assessment. It involves collecting information about a target system, such as publicly accessible services, open ports, service versions, and technologies used by a web application. This project was performed only against the authorized vulnerable training target provided for the internship.

2. Objective

- Identify publicly accessible services on the target.
- Discover open ports.
- Identify services and versions where possible.
- Determine technologies used by the web application.
- Document commands, results, and supporting screenshots.

3. Target Information

Target URL: http://testphp.vulnweb.com/

Target Type: Intentionally vulnerable web application used for authorized security training.

Testing Scope: Reconnaissance and service enumeration only.

4. Tools Used

Nmap: Used for port scanning, service detection, and basic enumeration.

WhatWeb: Used to identify technologies and web application components.

Netcat: Used for basic network/service connectivity checks where required.

5. Methodology

1. Confirm the authorized target.
2. Perform an initial Nmap scan to identify open ports.
3. Perform service/version detection using Nmap.
4. Use WhatWeb to identify web technologies.
5. Use Netcat only where a basic service connectivity check is useful.
6. Record the actual outputs and capture screenshots.
7. Analyze the exposed services and prepare recommendations.

6. Commands Used

Run the following against the authorized target and replace the placeholders with your actual outputs.

```
nmap testphp.vulnweb.com nmap -sV
testphp.vulnweb.com whatweb
http://testphp.vulnweb.com/
# Netcat: use only for a port you actually identified as open
nc -v testphp.vulnweb.com [OPEN_PORT]
```

Note: Do not copy sample scan results into this report. Paste only the results produced by your own authorized scan.

7. Results and Findings

Record the actual findings from your scans in the table below.

Port	Protocol	Service	Version	Status
80	TCP	http	Microsoft IIS httpd 8.5	Open (nmap -sV; OS: Windows)
80	TCP	tcpwrapped	Microsoft-IIS/8.5 (http-serverheader)	Open (nmap -sC sV)
80	TCP	http	Not identified (-script httptitle,http-headers run)	Filtered (observed in one scan)
Other 999 TCP ports	TCP	-	-	Filtered (no response)

Technology Identification:

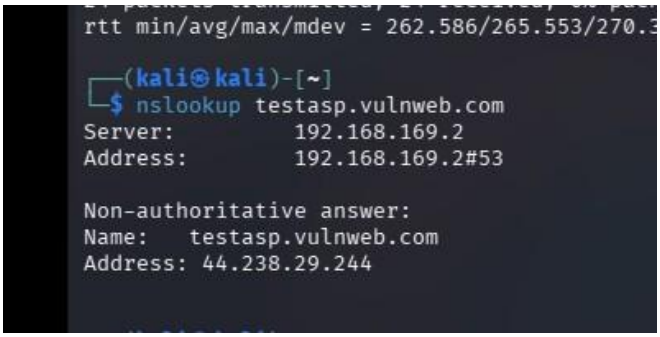
Technology / Component	Evidence from WhatWeb
Web server	HTTPServer[Microsoft-IIS/8.5], Microsoft-IIS[8.5]
Framework / platform	ASP_NET, X-Powered-By[ASP.NET]
Session cookie	Cookies[ASPSESSIONIDAADSRCSB]
Page title	Title[acuforum forums]
IP address / country	IP[44.238.29.244], Country[UNITED STATES][US]

Additional evidence: Netcat (nc -v testasp.vulnweb.com 80) confirmed port 80 open and a HEAD request returned HTTP/1.1 200 OK with Server: Microsoft-IIS/8.5, X-Powered-By: ASP.NET and an ASPSESSIONID cookie. DNS resolved testasp.vulnweb.com to 44.238.29.244 (reverse DNS: ec2-44-23829-244.us-west-2.compute.amazonaws.com), and ping replies showed TTL 128, consistent with a Windows host.

8. Screenshots / Evidence

Insert your own screenshots in this section. Suggested evidence:

- Figure 1: Target / initial reconnaissance command.
- Figure 2: Nmap port scan output.
- Figure 3: Nmap service/version detection output.
- Figure 4: WhatWeb technology identification output.
- Figure 5: Netcat connectivity check, if used.
- Figure 6: Final findings table or consolidated evidence.



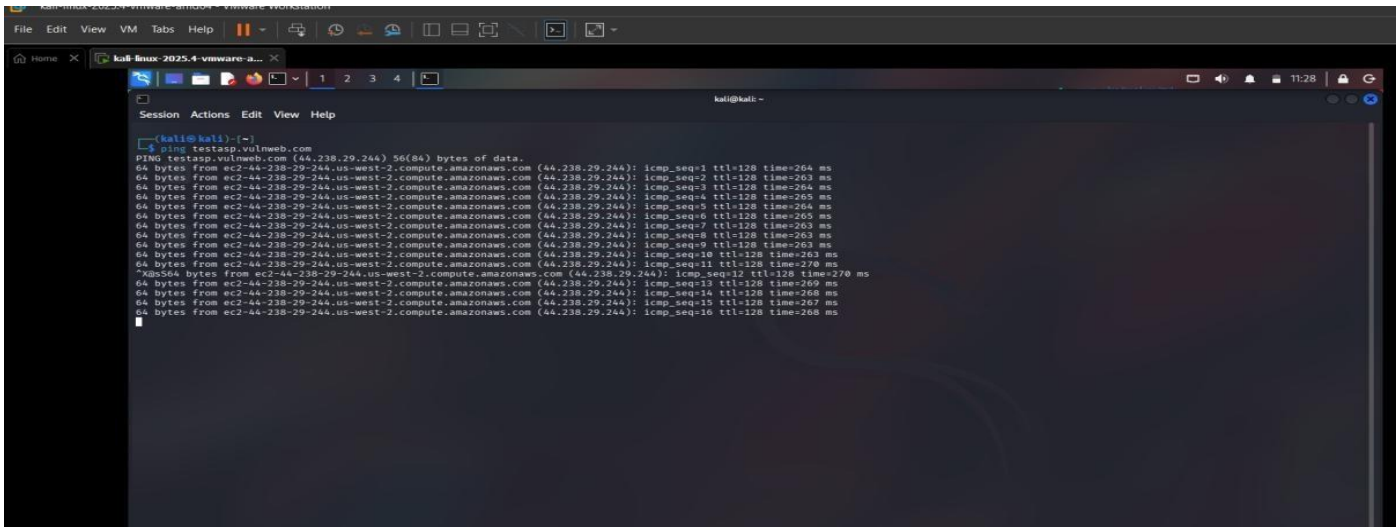


Figure 1 – Target DNS resolution (nslookup) and connectivity check (ping) for testasp.vulnweb.com.

```
┌──(kali㉿kali)-[~]
└─$ nmap 44.238.29.244
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-04 11:30 EDT
Stats: 0:00:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 73.50% done; ETC: 11:31 (0:00:12 remaining)
Stats: 0:00:50 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 76.10% done; ETC: 11:31 (0:00:16 remaining)
Stats: 0:01:20 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 79.77% done; ETC: 11:32 (0:00:20 remaining)
Nmap scan report for ec2-44-238-29-244.us-west-2.compute.amazonaws.com (44.238.29.244)
Host is up (0.050s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http
```

Figure 2 – Nmap port scan output (nmap 44.238.29.244).

```
Nmap done: 1 IP address (1 host up) scanned in 88.09 seconds

┌──(kali㉿kali)-[~]
└─$ nmap -sV testasp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-04 11:32 EDT
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 80.60% done; ETC: 11:32 (0:00:07 remaining)
Stats: 0:01:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.60% done; ETC: 11:33 (0:00:11 remaining)
Nmap scan report for testasp.vulnweb.com (44.238.29.244)
Host is up (0.013s latency).
rDNS record for 44.238.29.244: ec2-44-238-29-244.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
80/tcp open  http    Microsoft IIS httpd 8.5
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 92.46 seconds
```

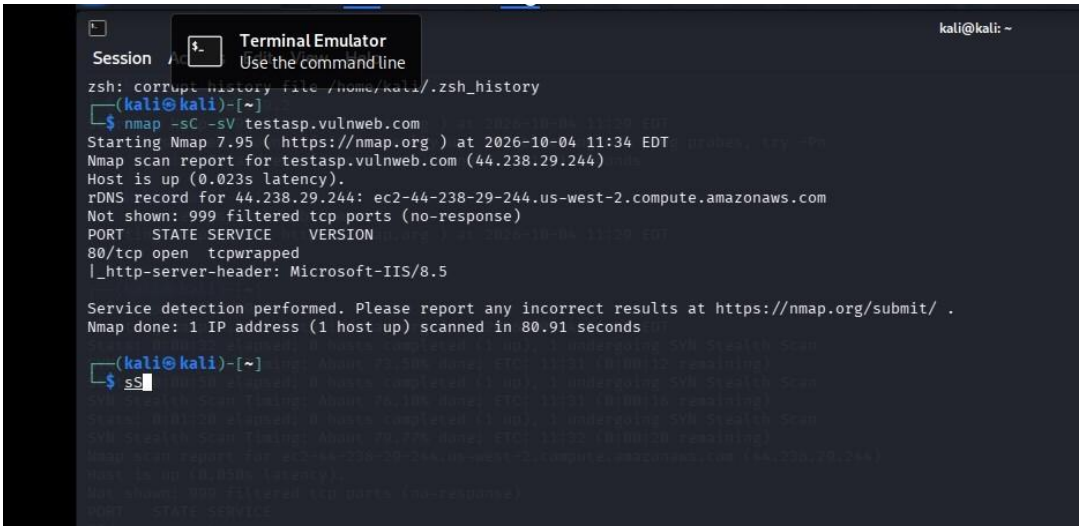


Figure 3 – Nmap service/version detection output (nmap -sV and nmap -sC -sV).



Figure 4 – WhatWeb technology identification output.



Figure 5 – Netcat connectivity check on port 80 (HEAD request).



Figure 6 – Additional Nmap check (nmap -p 80 --script http-title,http-headers).

9. Security Impact

Publicly accessible services increase the observable attack surface of a system. Unnecessary or outdated services may increase security risk if they are not properly configured, patched, monitored, or restricted. An open port by

itself is not necessarily a vulnerability; the actual risk depends on the service, version, configuration, exposure, and other security controls.

10. Recommendations

- Disable services that are not required.
- Restrict unnecessary network exposure using firewall or access-control rules.
- Keep exposed services and software updated.
- Monitor publicly accessible services and review logs.
- Use secure configurations and least-privilege principles.
- Perform periodic authorized security assessments.

11. Conclusion

This project provided practical experience in network reconnaissance and service enumeration. Nmap was used to identify open ports and services, while WhatWeb was used to identify web technologies. The collected evidence was documented through commands, scan results, and screenshots. The exercise demonstrates why reconnaissance is an important first step before a broader security assessment.

Final Checklist

☐ Name, email and registration number added

☐ Correct target mentioned

☐ Commands are the commands I actually used

☐ Actual Nmap output added

☐ Actual WhatWeb output added

☐ Netcat evidence added if used
☐ Screenshots are readable and numbered

☐ Findings table completed

☐ Recommendations added

☐ Conclusion added

☐ Report exported to PDF

☐ Project folder contains report and screenshots